

# Ogan K. Kumova, PhD

1005 1st St NE, Apt 1059 · Washington, DC 20002 · (609) 458-6761 · okumova@gmail.com

## PROFESSIONAL SUMMARY

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Science policy professional and immunologist with expertise bridging federal health policy and biomedical research. At the National Academies of Sciences, Engineering, and Medicine, manages expert consensus studies on vaccine safety commissioned by the CDC, VA, DoD, and FDA — overseeing the full lifecycle from expert recruitment through Congressional briefing. Contributing Editor on two major federal reports on vaccine safety and serves as a trusted scientific voice for policymakers and public health officials. PhD in Microbiology & Immunology with postdoctoral research at the FDA. Bilingual (Turkish/English). Fluent in the scientific, regulatory, and political dimensions of public health.

## CORE COMPETENCIES

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Expert Committee Management · Evidence Synthesis & Scientific Review · Federal Policy Analysis · Stakeholder Coordination (CDC, FDA, VA, DoD)

Congressional Communication · Scientific Writing & Editing · Report Production & Dissemination · Budget & Contract Management

Vaccine Safety & Immunology · Public Health Communication · Cross-Sector Collaboration · Program & Project Management

## PROFESSIONAL EXPERIENCE

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### Program Officer, Vaccine Safety

*February 2023 – Present*

*National Academies of Sciences, Engineering, and Medicine — Washington, D.C.*

- Manage expert committee consensus studies on vaccine safety and immunology for federal agency sponsors (VA, DoD, CDC, FDA), overseeing the full study lifecycle: expert recruitment, committee coordination, meeting logistics, evidence synthesis, and report production.
- Currently supporting two active ad hoc studies: (1) Neurodegenerative Outcomes and Selected Military Exposures; and (2) Examination of the Direct Neurocognitive Effects of SARS-CoV-2 Infection Across the Lifespan and the Neuroprotective Effects of COVID Vaccination.
- Served as Contributing Editor on two published National Academies reports: Evidence Review of the Adverse Effects of COVID-19 Vaccination and Intramuscular Vaccine Administration (National Academies Press, 2024) and Vaccine Risk Monitoring and Evaluation at the Centers for Disease Control and Prevention (National Academies Press, 2025).
- Delivered invited briefing on COVID-19 vaccine safety to Congressional staff of the U.S. House and Senate Appropriations Committees and the Senate HELP Committee (October 2025).
- Manage project timelines, vendor contracts, and budget tracking across concurrent study portfolios; coordinates multi-stakeholder processes across federal sponsors, committee members, external reviewers, and institutional staff.

### ORISE Postdoctoral Research Fellow

*October 2020 – February 2023*

*U.S. Food and Drug Administration, Office of Vaccines Research and Review — Silver Spring, MD · Supervisor:*

*Margaret C. Bash, MD, MPH · NIH-U19 Gonorrhea Vaccine Cooperative Research Center*

- Conducted independent research within the NIH-U19 Gonorrhea Vaccine Cooperative Research Center — a multi-institutional collaboration across FDA, academic, and NIH sites — developing liposome- and OMV-based *Neisseria gonorrhoeae* vaccine candidates.
- Gained direct exposure to the FDA regulatory science environment: GLP standards, SOP development, regulatory documentation, and OVR's preclinical vaccine safety evaluation framework.
- Contributed to grant progress reports, multi-institutional data-sharing, and consortium coordination across federal and academic partners.
- Results published in *Journal of Infectious Diseases* (2025; PMID 40971931).

## Laboratory Manager

December 2014 – August 2015

*Drexel University College of Medicine — Philadelphia, PA*

- Managed daily operations, NIH grant budget, regulatory compliance (IACUC, IBC), and personnel coordination for an active immunology research laboratory.
- Wrote and maintained protocols for experimental procedures, regulatory submissions, and safety documentation.

## Laboratory Technician II

December 2012 – December 2014

*Drexel University College of Medicine — Philadelphia, PA*

- Supported research in an active infectious disease laboratory investigating *Legionella pneumophila* pathogenesis; contributed to three peer-reviewed publications.

## EDUCATION

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### Doctor of Philosophy, Microbiology & Immunology

2015 – 2020

*Drexel University College of Medicine — Philadelphia, PA*

GPA: 4.0 · Dissertation: Altered Innate Immune Responses to Influenza in the Neonate · Minor: Health Administration

### Master of Science, Infectious Diseases

2013 – 2015

*Drexel University College of Medicine — Philadelphia, PA*

GPA: 3.97

### B.S. in Biochemistry & B.S. in Bioinformatics

2005 – 2009

*University of the Sciences in Philadelphia — Philadelphia, PA*

Minor in Mathematics

## POLICY & EDITORIAL OUTPUTS

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- Ancona D, Kumova OK, Stratton K, Henney JE, editors. Vaccine Risk Monitoring and Evaluation at the Centers for Disease Control and Prevention. Washington, DC: National Academies Press; 2025. doi:10.17226/29240. PMID: 41289413.
- Rosenberg D, Kumova OK, Stratton K, Bass AR, editors. Evidence Review of the Adverse Effects of COVID-19 Vaccination and Intramuscular Vaccine Administration. Washington, DC: National Academies Press; 2024. doi:10.17226/27746. PMID: 39312602.
- Congressional Briefing: Vaccine Risk Monitoring and Evaluation at the CDC — for staff of the U.S. House and Senate Appropriations Committees and the Senate HELP Committee. October 2025.
- Congressional Briefing: COVID-19 Vaccine Safety — for staff of the U.S. House and Senate Appropriations Committees and the Senate HELP Committee. May 2024.

## SELECTED PEER-REVIEWED PUBLICATIONS

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15 peer-reviewed articles total. Selected publications most relevant to vaccine safety and immunology:

- Rao AS, Onufer AP, **Kumova O**, Roberts LTN, Mossabeh R, Pascasio J, Carey AJ. Myeloid TLR2 signaling amplifies immunopathology in influenza-infected murine neonates. *Am J Respir Cell Mol Biol*. 2026;aanag050. PMID: 42089286.
- Matthias KA, Connolly KL, Jerse AE, **Kumova OK**, Macintyre AN, Gray MC, Thomas KS, Criss AK, Zielke RA, Li L, Sikora AE, Ramirez-Bencomo F, Thistlethwaite A, Derrick JP, Lu WE, Bash MC. Evaluation of immunization route in induction of vaccine-mediated anti-gonococcal immune responses in a murine model of ascending infection. *J Infect Dis*. 2025;232(5):e765–e777. PMID: 40971931.
- Rao AS, Ugwu N, Onufer AP, **Kumova O**, Carey AJ. Cathelicidin-related antimicrobial peptide (CRAMP) is toxic during neonatal murine influenza virus infection. *J Immunol*. 2025;214(5):1022–1031. PMID: 40101750.
- Kumova OK**, Galani IE, Rao A, Johnson H, Triantafyllia V, Matt SM, Pascasio J, Gaskill PJ, Andreakos E, Katsikis PD, Carey AJ. Severity of neonatal influenza infection is driven by type I interferon and oxidative stress. *Mucosal Immunol*. 2022;15(6):1309–1320. PMID: 36352099.

Gary E, O'Connor M, Chakhtoura M, Tardif V, **Kumova OK**, Malherbe DC, Sutton WF, Haigwood NL, Kutzler MA, Haddad EK. Adenosine deaminase-1 enhances germinal center formation and functional antibody responses to HIV-1 envelope DNA and protein vaccines. *Vaccine*. 2020;38(22):3821–3831. PMID: 32280045.

**Kumova OK**, Fike AJ, Thayer JL, Nguyen LT, Mell JC, Pascasio J, Stairiker C, Leon LG, Katsikis PD, Carey AJ. Lung transcriptional unresponsiveness and loss of early influenza virus control in infected neonates is prevented by intranasal *Lactobacillus rhamnosus* GG. *PLoS Pathog*. 2019;15(10):e1008072. PMID: 31603951.

#### ***National Academies Reports — Contributing Editor***

Ancona D, **Kumova OK**, Stratton K, Henney JE, editors. Vaccine Risk Monitoring and Evaluation at the Centers for Disease Control and Prevention. Washington, DC: National Academies Press; 2025. doi:10.17226/29240. PMID: 41289413.

Rosenberg D, **Kumova OK**, Stratton K, Bass AR, editors. Evidence Review of the Adverse Effects of COVID-19 Vaccination and Intramuscular Vaccine Administration. Washington, DC: National Academies Press; 2024. doi:10.17226/27746. PMID: 39312602.

### **ADDITIONAL EXPERIENCE**

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#### **Senior Adjunct Faculty**

*August 2020 – Present*

*Rowan University / Rowan College at Burlington County — STEM Division*

- Designs and leads courses in microbiology, biology, and health sciences for pre-professional and non-traditional students; known for making technically demanding material accessible and engaging across in-person, hybrid, and fully online formats.
- Recognized for teaching excellence: Excellence in Teaching Award, 2022 & 2023.

#### **Lecturer & Visiting Faculty**

*2015 – Present*

*Drexel University College of Medicine · Uniformed Services University Medical College*

- Delivers invited graduate and medical school lectures on vaccine development, immunology, and molecular pathogenesis to PhD students, MD students, and military physicians and scientists.

#### **Museum Educator, Program Developer & Acting Camp-In Director**

*May 2003 – March 2020*

*Franklin Institute Science Museum — Philadelphia, PA*

- Delivered live science programming to diverse public audiences; developed K–12 workshop curricula and traveling programs, building extensive experience in translating complex science across age groups and backgrounds.
- Served as Acting Director of the overnight camp-in program, managing staff, budgets, and events of up to 400 participants per event; contributed to the Philadelphia Science Festival.

### **SKILLS & LANGUAGES**

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**Policy & Management:** Scientific Writing & Editing · Evidence Synthesis & Systematic Review · Federal Grant & Budget Management · Stakeholder & Vendor Coordination

**Scientific Expertise:** Vaccine Immunology · Infectious Disease · Public Health · Regulatory Science (FDA/GLP)

**Digital Tools:** Blackboard · Canvas · Kaltura · Microsoft Office · LIMS/Quartzy · AI-assisted research tools

**Languages & Certifications:** Turkish (native) · English (native) · CRLA Level II Certified Tutor · First Aid & CPR